Spherepop as an Entropy-Decreasing Rewriting Category: Axiomatic Foundations, Functorial Geometry, and the RSVP Embedding

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Abstract

We present a unified axiomatic treatment of the Spherepop framework, drawing on its instantiations as an event-history calculus, a typed functional language (SPC), an OS kernel semantics, and a geometric theory of coherence within the Relativistic Scalar-Vector Plenum (RSVP). The paper proceeds in four movements. First, we formalize the *event-history calculus* through five axioms—sequential irreversibility, entropy monotonicity, a quartet of primitive morphisms (Pop, Refuse, Bind, Collapse), authority of history, and a collapse-induced identity relation—and establish that the resulting category **SP** carries the structure of a strict monoidal category with an entropy functional. Second, we connect this to the *Spherepop Calculus* (SPC), showing that SPC's typed primitives (Sphere, Pop, Merge, Choice) correspond precisely to the generating morphisms of **SP** interpreted in a presheaf topos enriched with the Giry distribution monad. Third, we construct a *geometric realization functor* $F : \mathbf{SP} \to \mathbf{RSVP}$ mapping option-spaces to smooth manifolds carrying scalar, vector, and entropy fields, and translating each primitive morphism into a localized field perturbation—option-elimination becomes boundary sharpening, dependency binding becomes directed vector-field coupling, and quotient collapse becomes Wilsonian renormalization. Fourth, we characterize the deep structural asymmetry between the two frameworks: Spherepop *accumulates* constraint in a log-bound combinatorial system; RSVP *redistributes* coherence through continuous field dynamics. We conclude with open

problems in faithfulness, non-trivial entropy functionals for Bind, and compact-closed completions.

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1 Introduction

1.1 Background and Motivation

The Spherepop framework has developed along four distinct but convergent trajectories. As an *event-history calculus*, it proposes that worlds are constructed through irreversible commitment rather than state update: the fundamental ontological unit is not a snapshot but a trajectory, the accumulated result of events that permanently constrain the futures remaining accessible. As a *typed functional language*, the Spherepop Calculus (SPC) offers a geometric reinterpretation of λ-abstraction as nested spherical scoping, extending the Calculus of Constructions with categorical parallelism (Merge as tensor) and compositional probability (Choice via the distribution monad). As an *OS kernel semantics*, it disciplines utility programs as structured consumers of authoritative event logs, enforcing deterministic replay, ABI stability, and strict separation between kernel state and derived views. As a *geometric theory of coherence*, it situates all of these within the RSVP plenum, where a scalar coherence potential Φ, a velocity field v, and an entropy density S evolve under coupled PDEs whose large-scale behavior is the thermodynamic limit of Spherepop's discrete dynamics.

The present paper provides, for the first time, a unified formal treatment connecting all four trajectories. The central claim is:

> *Spherepop is a finitely generated entropy-decreasing rewriting category; RSVP is the smooth dynamical realization of its morphisms as coherence flows in a plenum, via a geometric realization functor that translates option-elimination into boundary sharpening, dependency into vector-field coupling, and collapse into renormalization.*

This is not a metaphor. The functor $F : \mathbf{SP} \to \mathbf{RSVP}$ is constructed explicitly, object by object and generating morphism by generating morphism. Its existence and properties are stated as a theorem (Theorem 6.3), with proof sketches that identify exactly which mathematical facts are fully established and which require additional modeling choices.

1.2 Irreversibility as Semantic Axiom

The philosophical core of Spherepop is the elevation of thermodynamic irreversibility from implementation artifact to semantic axiom. The basic claim is that any system that performs non-trivial computation in the presence of finite resources must dissipate information and lose degrees of freedom [Landauer, 1961, Bennett, 1973]. Rather than treating this as a limitation, Spherepop makes it constitutive: meaning arises not from instantaneous configuration but from the irreversible sequence of events through which a system has passed.

This positions Spherepop in sharp contrast with state-based computational paradigms. Classical process calculi permit reversal; cybernetic control theories treat adaptation as regulation over an invariant possibility space; probabilistic programming languages sample from distributions without accumulating causal history. Spherepop replaces all of these with a single organizing principle: *to choose is to permanently reshape the topology of the future.*

1.3 Structure of the Paper

Section 2 establishes the primitive data and categorical structure of **SP**. Section 3 presents the five formal axioms. Section 4 connects the event-history calculus to the Spherepop Calculus and its presheaf topos semantics. Section 5 discusses the OS kernel interpretation and its structural implications. Section 6 constructs the functor $F : \mathbf{SP} \to \mathbf{RSVP}$. Section 7 provides the structural comparison. Section 8 treats the discrete Lagrangian mechanics of commitment. Section 9

develops the sheaf-theoretic interpretation of worldhood and the meld operator. Section 10 surveys mathematical status and open problems. Section 11 compresses the paper to its minimal core.

2 Primitive Data and the Category SP

2.1 Option-Spaces

Definition 2.1 (Option-space)**.** An *option-space* is a pair $(\Omega, \mathcal{A})$ where Ω is a non-empty set (the set of admissible futures) and $\mathcal{A} \subseteq 2^{\Omega}$ is a distinguished family closed under finite intersection, representing the *admissible subfutures* at any stage of commitment.

The intuition is concrete: Ω is the set of everything that could still happen from the current historical position; $\mathcal{A}$ encodes the structural dependencies among those futures. A pop shrinks Ω; a bind reduces $\mathcal{A}$ without changing Ω; a collapse quotients both.

Definition 2.2 (Entropy functional)**.** A *Spherepop entropy functional* is a map

$$H : \mathrm{Ob}(\mathbf{SP}) \longrightarrow \mathbb{R}_{\geq 0}$$

satisfying: (i) $H(\Omega) = 0$ iff $|\Omega| = 1$; (ii) H is monotone in the sense that $\Omega' \subseteq \Omega \Rightarrow H(\Omega') \leq H(\Omega)$.

The Shannon entropy $H(\Omega) = -\sum_{x \in \Omega} p(x) \log p(x)$ (for any probability measure on Ω) provides the canonical instance, but the axioms hold for any monotone subadditive functional, including topological entropy on the admissibility lattice.

2.2 Morphisms and Composition

The category **SP** has option-spaces as objects. A morphism $e : \Omega \to \Omega'$ is an *irreversible event*—a structure-preserving map that weakly decreases entropy. Morphisms compose associatively; the identity at Ω is the degenerate history id_{Ω} that leaves all structure unchanged.

A *history of length n* is a composable sequence

$$h = e_n \circ e_{n-1} \circ \cdots \circ e_1 : \Omega_0 \longrightarrow \Omega_n.$$

Histories are the primary semantic objects of **SP**. Because morphisms are irreversible, **SP** is not a groupoid: no non-identity morphism has an inverse. This asymmetry encodes the arrow of time directly into the algebra.

3 The Five Axioms of Spherepop

Axiom 1 (Sequential Irreversibility)**.** Composition is strictly associative. There is a distinguished initial object Ω_0. There is no morphism inverse to any non-identity event: for all $e : \Omega \to \Omega'$ with $e \neq \mathrm{id}_\Omega$, there is no $e^{-1} : \Omega' \to \Omega$ in **SP**.

Axiom 2 (Entropy Monotonicity)**.** For every morphism $e : \Omega \to \Omega'$ in **SP**:

$$H(\Omega') \leq H(\Omega).$$

Strict inequality holds for Pop and Ref. Bind preserves H globally while reducing structural symmetry. Col may produce non-strict inequality when a quotient is entropy-neutral at coarse scale but eliminates fine structure.

Axiom 3 (Primitive Morphisms)**.** The category **SP** is generated by exactly four families:

(i) **Pop** ($\mathrm{Pop} : \Omega \to \Omega'$). Strict subobject inclusion in the reverse direction:

$$\Omega' \subsetneq \Omega, \qquad H(\Omega') < H(\Omega).$$

One or more options are permanently excluded. This is the fundamental operation of world construction: to pop is to actualize one future by eliminating others.

(ii) **Refuse** ($\mathrm{Ref} : \Omega \to \Omega'$). Geometrically identical to Pop—same subobject inclusion—but carrying a semantic tag distinguishing deliberate non-selection from positive selection. Formally, Ref and Pop are distinct morphisms over the same underlying set map; the distinction is tracked by an auxiliary accounting functor $\mathcal{A} : \mathbf{SP} \to \mathbf{C}$ and matters for normative interpretation.

(iii) **Bind** ($\mathrm{Bind} : \Omega \to \Omega$). An endomorphism reducing the admissible family:

$$\mathcal{A}_{\mathrm{Bind}(\Omega)} \subsetneq \mathcal{A}_\Omega,$$

without removing elements from Ω. Binding introduces a dependency constraint—a partial ordering or precedence among options—without yet elim-

inating them. It corresponds to the introduction of structure rather than reduction of space.

(iv) **Collapse** ($\mathrm{Col} : \Omega \to \Omega/{\sim}$). A quotient morphism induced by an equivalence relation $\sim$ on Ω, required to preserve causal structure:

$$x \sim y \implies \text{the causal past of } x = \text{the causal past of } y.$$

Collapse reduces descriptive complexity while preserving causal consequence. Two futures differing only along dimensions irrelevant to downstream commitments are identified. This is the only operator permitted to reduce accumulated action.

Remark 3.1. The four generators correspond exactly to the SPC primitives and OS kernel event types. In SPC: Pop $\leftrightarrow$ $\mathrm{Pop}(f, u)$ (application), Bind $\leftrightarrow$ $\mathrm{Sphere}(x : A.\, t)$ (abstraction creating dependency), Col $\leftrightarrow$ Merge (identification under equivalence). In the OS kernel: Pop $\leftrightarrow$ `POP`, Bind $\leftrightarrow$ `LINK`, Col $\leftrightarrow$ `COLLAPSE`, Ref $\leftrightarrow$ `UNLINK` with accounting tag.

Axiom 4 (Authority of History)**.** The state at time n is defined solely by replay:

$$\Omega_n = (e_n \circ \cdots \circ e_1)(\Omega_0).$$

There is no hidden state. Any two processes with access to the same event log and the same Ω_0 reconstruct the same Ω_n. The category **SP** admits no monad structure that could introduce effectful side-channels.

This axiom is structurally equivalent to the event-sourcing principle in distributed systems: *state = replay(log)*. It is also the basis for the OS kernel constraint that utilities may not mutate kernel state directly.

Axiom 5 (Collapse Identity)**.** Let q be a collapse policy (a family of equivalence relations parameterized by resolution level). Two histories h, h' are *q-equivalent*, written $h \sim_q h'$, if they differ only by refinements annihilated by q:

$$h \sim_q h' \iff q\Big(h(\Omega_0)\Big) = q\Big(h'(\Omega_0)\Big).$$

Identity in **SP** is well-defined modulo q.

3.1 Structural Consequences

Proposition 3.2 (Strict Monoidal Structure)**.** $(\mathbf{SP}, \circ, \mathrm{id}_{\Omega_0})$ *is a strict monoidal category in which the entropy functional H constitutes a strict monoidal functor $H : \mathbf{SP} \to (\mathbb{R}_{\geq 0}, +, 0)$.*

Proof sketch. Associativity and unit laws follow from Axiom 1. Strictness holds because composition is defined as list concatenation on event sequences, which is strictly associative. Functoriality of H follows from Axiom 2: $H(h \circ h') \leq H(h') \leq H(\Omega_0)$, and $H(\mathrm{id}) = H(\Omega_0)$. □

Proposition 3.3 (Universal Property of Collapse)**.** Col *is the coequalizer of all pairs of morphisms in* **SP** *that produce the same causal history on Ω_0. For any functor $G : \mathbf{SP} \to \mathcal{C}$ such that $G(e) = G(e')$ whenever $e \sim_q e'$, there is a unique factorization through* Col.

Proof sketch. Standard coequalizer construction in a category with quotient objects. The causal structure preservation clause of Axiom 3(iv) ensures that the induced quotient morphism is well-defined. □

Proposition 3.4 (Meld as Pushout)**.** *The* meld *operator* $\mathrm{Meld}_\pi(H_1, H_2) = H_3$ *is the pushout of two histories over their common prefix, modulo the preference policy π:*

$$\begin{tikzcd} \Omega_0 \arrow[r, "H_1"] \arrow[d, "H_2"'] & \Omega_1 \arrow[d] \\ \Omega_2 \arrow[r] & \Omega_3 = \Omega_1 \sqcup_{\Omega_0} \Omega_2/{\sim_\pi} \end{tikzcd}$$

The policy π determines which incompatible commitments across Ω_1 and Ω_2 are identified (collapsed) and which are preserved. This decomposes meld into a collapse step followed by a categorical gluing: $\mathrm{Meld}_\pi \equiv \mathrm{glue} \circ \mathrm{Col}_\pi$.

4 The Spherepop Calculus as Typed Instantiation

The Spherepop Calculus (SPC) provides the type-theoretic instantiation of **SP**. Its grammar of terms is:

$$t, u \ ::= \ x \ \mid \ a \ \mid \ \mathrm{Sphere}(x{:}A.\,t) \ \mid \ \mathrm{Pop}(t, u) \ \mid \ \mathrm{Merge}(t, u) \ \mid \ \mathrm{Choice}(p, t, u),$$

and the core reduction rules are:

$$\begin{aligned}
\mathrm{Pop}(\mathrm{Sphere}(x\colon A.\, t),\ u) &\to t[u/x] && (\beta\text{-reduction})\\
\mathrm{Choice}(p, t, u) &\to \begin{cases} t & \text{w.p. } p \\ u & \text{w.p. } 1-p \end{cases} && (\text{probabilistic sampling})\\
\mathrm{Merge}(t, u) &\equiv \mathrm{Merge}(u, t) && (\text{commutativity})
\end{aligned}$$

4.1 Correspondence with SP Generators

The four SPC primitives correspond to the generators of **SP** as follows:

SPC Primitive	**SP Generator**	**Action on Ω**
$\mathrm{Pop}(f, u)$	Pop/Ref	Strict subobject inclusion
$\mathrm{Sphere}(x\colon A.\, t)$	Bind	Dependency constraint, endomorphism
$\mathrm{Merge}(t, u)$	Col (pushout)	Quotient/identification
$\mathrm{Choice}(p, t, u)$	Pop + probabilistic tag	Weighted option exclusion

4.2 Presheaf Topos Semantics

SPC is interpreted in the presheaf topos $\mathbf{E} = [\mathbf{Sphere}^{\mathrm{op}}, \mathbf{Set}]$ enriched with the Giry distribution monad Dist [Giry, 1982]. The core semantic clauses are:

$$\mathrm{Sphere}(x\colon A.\, t) = \text{exponential object } B^A \text{ in } \mathbf{E} \tag{1}$$

$$\mathrm{Pop}(t, u) = \mathrm{ev} \circ \langle t, u \rangle \tag{2}$$

$$\mathrm{Merge}(t, u) = \langle t, u \rangle : \Gamma \to A \otimes A \tag{3}$$

$$\mathrm{Choice}(p, t, u) = \lambda\gamma.\ p\, \delta_{t(\gamma)} + (1-p)\, \delta_{u(\gamma)} \tag{4}$$

This gives Merge the structure of a symmetric monoidal tensor and Choice the structure of a convex algebra. The Independent Channels Lemma follows directly from independence of the product measure:

$$\Pr\Big[\mathrm{FoldOr}(\mathrm{Merge}(R_1, \ldots, R_n)) = \mathrm{Doom}\Big] = 1 - \prod_{i=1}^{n} (1 - p_i).$$

4.3 Meta-Theory

SPC satisfies the standard type-safety results (Preservation and Progress) together with Adequacy of the probabilistic semantics, whose precise statement is: for any closed term t of type A and measurable $h : A \to \mathbb{R}$,

$$\mathbb{E}_{\text{operational}}[h] = \mathbb{E}_t[h].$$

The connection to **SP** makes this also a statement about entropy: the entropy functional H applied to the final option-space $F^{-1}(t)$ coincides, up to units, with the entropy density $\int_M S\, d\text{vol}$ in the RSVP realization (Theorem 6.3(c)).

5 The OS Kernel Interpretation

The Spherepop OS kernel instantiates **SP** as an *event-sourced append-only log system*. The structural correspondences are exact:

SP Structure	**OS Kernel Counterpart**
History $h = e_n \circ \cdots \circ e_1$	Append-only event log
Axiom 4 (Authority of History)	State = replay(log)
Bind (endomorphism)	`LINK` event
Col (quotient)	`COLLAPSE` + snapshot
Pop / Ref	`POP` / `UNLINK` with accounting tag
View generator (utility)	Non-authoritative derived view
Overlay manager	Speculative branch outside kernel

The design constraint that utilities may not mutate kernel state directly (Authoritative Discipline is precisely Axiom 4: there is no hidden state, and all authoritative changes must be expressed as proposed events submitted to the arbiter. The prohibition on time-dependent behavior and hidden mutable caches (Non-Goals) is a consequence of the same axiom.

The *view-cause separation* enforced by the kernel—views may be dropped or recomputed without semantic consequence—corresponds categorically to the existence of the accounting functor $\mathcal{A} : \mathbf{SP} \to \mathbf{C}$ that tracks Ref vs. Pop distinctions without affecting the underlying morphism in $\mathcal{O}$.

6 RSVP as a Geometric Realization Functor

6.1 The RSVP Plenum

The Relativistic Scalar-Vector Plenum is a smooth field theory on a Riemannian manifold M carrying three coupled fields: a scalar coherence potential $\Phi \in C^\infty(M)$, a velocity field $v \in \mathfrak{X}(M)$, and an entropy density $S \in C^\infty(M, \mathbb{R}_{\geq 0})$. Their dynamics are governed by the entropy-transport equation

$$\partial_t S = \nabla \cdot (\kappa \, \nabla \Phi), \tag{5}$$

together with continuity equations for v and conservation of total entropy flux [Jacobson, 1995, Verlinde, 2011].

A *pop derivative* along a spherical embedding $f : \mathrm{Sphr} \hookrightarrow M$ is defined as the curvature-weighted radial variation of Φ:

$$\partial_P \Phi = \lim_{\epsilon \to 0} \frac{\Phi(\mathrm{Sphr} + \epsilon) - \Phi(\mathrm{Sphr})}{\epsilon} = (\nabla \cdot \mathbf{n}) \, \partial_r \Phi + v \cdot \nabla \Phi. \tag{6}$$

Positive $\partial_P \Phi$ indicates emergent coherence; negative values signal dissolution.

6.2 The Category RSVP

Definition 6.1 (RSVP objects and morphisms)**.** An *RSVP state* is a quadruple (M, Φ, v, S) as above. A morphism $(M, \Phi, v, S) \to (M', \Phi', v', S')$ is a smooth map $\varphi : M \to M'$ satisfying: (i) $\varphi^* S' \geq S$ (entropy non-decrease); (ii) the entropy flux $\kappa \nabla \Phi$ is preserved up to boundary terms under φ_*.

6.3 Construction of F

Definition 6.2 (Geometric realization functor)**.** Define $F : \mathbf{SP} \to \mathbf{RSVP}$ as follows.

On objects. Given $\Omega \in \mathrm{Ob}(\mathbf{SP})$, set

$$F(\Omega) = \Big(\Delta(\Omega), \ \Phi_\Omega, \ v_\Omega, \ S_\Omega \Big)$$

where $\Delta(\Omega)$ is the probability simplex over Ω (or a smooth approximation), S_Ω is the Shannon entropy density, and Φ_Ω, v_Ω are initialized as smooth interpolants of the admissibility structure $\mathcal{A}$.

On generating morphisms. Each primitive event is realized as a localized smooth perturbation:

Pop under F.

$$F(\text{Pop}) : (M, \Phi, v, S) \longmapsto (M \setminus U, \ \ \Phi + \delta\Phi_\partial, \ \ v, \ \ S + \delta S)$$

where $U \subset M$ is the open set corresponding to eliminated options, $\delta\Phi_\partial > 0$ is a boundary-sharpening increment concentrated near ∂U, and $\delta S \geq 0$ enforces Axiom 2. Geometrically: option-elimination $\leftrightarrow$ localized entropy-gradient increase + boundary sharpening. This is the field-theoretic realization of the pop derivative (6): positive $\partial_P \Phi$ marks the contraction of the admissible manifold.

Refuse under F. Identical field perturbation to $F(\text{Pop})$, but the perturbation carries a signed tag in a distinguished cohomological degree, distinguishing it from positive selection. The accounting functor $\mathcal{A}$ tracks this distinction without affecting the underlying field equations.

Bind under F.

$$F(\text{Bind}) : v \longmapsto v + \delta v_{ij}$$

where δv_{ij} is a directed coupling term between regions $i, j \subset M$ corresponding to the constrained options. Dependency binding $\leftrightarrow$ directed vector-field coupling. Entropy is globally preserved ($\delta S = 0$) but the symmetry group of (M, v) is reduced—exactly the content of Axiom 3(iii).

Collapse under F.

$$F(\text{Col}) : (M, \Phi, v, S) \longmapsto (\bar{M}, \ \bar{\Phi}, \ \bar{v}, \ \bar{S})$$

where $(\bar{M}, \bar{\Phi}, \bar{v}, \bar{S})$ is the Wilsonian renormalization-group flow image: high-frequency modes of Φ and v are integrated out, $\bar{M}$ is a coarser-resolution version of M, and $\bar{S}$ is the coarse-grained entropy. The action reduction formula of the discrete mechanics (Section 8) has its RSVP counterpart in the suppression of short-distance entropy fluctuations.

6.4 Functoriality and Entropy Compatibility

Theorem 6.3 (Functoriality of F)**.** *$F : \mathbf{SP} \to \mathbf{RSVP}$ is a well-defined functor. Specifically:*

(a) $F(\text{id}_\Omega) = \text{id}_{F(\Omega)}$*;*

(b) $F(e_2 \circ e_1) = F(e_2) \circ F(e_1)$*;*

(c) Entropy compatibility: $H(\Omega) = \int_{\Delta(\Omega)} S_\Omega \, \mathrm{dvol}$ *up to units, so entropy decrease in* **SP** *corresponds to net entropy production localized to* $\partial M'$ *in* **RSVP***.*

Proof sketch. (a) The identity event is the empty perturbation; the corresponding smooth map is $\mathrm{id}_{F(\Omega)}$. (b) Sequential perturbations on nested or disjoint regions of M compose as smooth maps. The locality of each perturbation (concentrated near ∂U for Pop, along δv_{ij} for Bind, in the RG flow for Col) ensures that $F(e_2) \circ F(e_1)$ has the same support as $F(e_2 \circ e_1)$. (c) By construction, $S_\Omega = -p_\Omega \log p_\Omega$ on $\Delta(\Omega)$, so $\int S \, d\mathrm{vol} = H(\Omega)$ by definition of the simplex measure. □

Commutative diagram.

$$\begin{tikzcd}
\Omega \arrow[r, "e"] \arrow[d, "F"'] & \Omega' \arrow[d, "F"] \\
(M, \Phi, v, S) \arrow[r, "F(e)"'] & (M', \Phi', v', S')
\end{tikzcd}$$

Merge product under F**.** The merge product μ°—the coherence-preserving convolution of two overlapping pop fields—corresponds under F to the RSVP-theoretic image of Meld_π (Proposition 3.4):

$$F(\mathrm{Meld}_\pi(H_1, H_2)) = \mu^\circ\Big(F(H_1), F(H_2)\Big) = \int_{\mathrm{Sph}_{12}} w(x, y) \, \Phi_1(y) \, \Phi_2(y) \, dV_y,$$

where the kernel w respects entropy flux and the categorical monoidal structure of μ°.

7 Structural Comparison

7.1 Not Layers, Not Metaphors: A Precise Duality

The relationship between **SP** and **RSVP** is often mischaracterized as a "levels of description" or "dual perspective" relationship. Both phrases suggest a symmetry that does not exist. The correct characterization is:

Spherepop (SP)	**RSVP**
Rewriting system	Thermodynamic limit of that system
Discrete computation	Continuous field theory
Category of histories	Derived geometric space
Constraint *accumulation*	Coherence *redistribution*
Log-bound, combinatorial	Field-bound, differential
Monotone, irreversible	Diffusive, dissipative
Col = coequalizer	$F(\mathrm{Col})$ = renormalization

Spherepop is the *algebra*; RSVP is its *analytic completion*. The functor F is the realization map between them, analogous to the relationship between a simplicial set and its geometric realization as a topological space, or between a rewriting system and its word problem in the limit.

7.2 Mapping to Standard Mathematical Frameworks

Category Theory.

- **SP** is a category of entropy-decreasing endofunctors on a poset of option-spaces.
- Col is the coequalizer of $\sim_q$-equivalent morphisms (Proposition 3.3).
- Meld is the pushout of two Bind-derived histories over a shared prefix (Proposition 3.4).
- Bind enriches hom-sets via a constraint relation, making **SP** an enriched category over the lattice of admissibility families.

Information Theory. The generating morphisms map to standard information-theoretic operations:

$$\text{Pop}: \quad H_{t+1} = H_t - \Delta, \quad \Delta > 0 \tag{7}$$

$$\text{Bind}: \quad I(X;Y) \text{ increases (mutual information coupling)} \tag{8}$$

$$\text{Col}: \quad \mathrm{K}(\Omega/\sim) \leq \mathrm{K}(\Omega), \quad \text{causal graph preserved} \tag{9}$$

The governing PDE on the RSVP side is the entropy-transport equation (5).

Distributed Systems. Spherepop is structurally isomorphic to the event-sourced architecture pattern: append-only log, state = replay(log), no hidden mutation, derived views allowed. Col corresponds to a snapshot: a compressed representation valid because of the causal preservation clause in Axiom 3(iv).

Cognitive Modeling. The endofunctor characterization frames a Spherepop agent as $A : \mathcal{O} \to \mathcal{O}$, preserving inclusion but not reversibility. Parsing becomes commitment (incoming structure is parsed into constraint rather than representation); identity follows from historical effect, not internal state. This supports the *log-bound process* account of agency: memory is accumulated constraint, not a store of representations.

8 The Mechanics of Commitment

Spherepop provides a discrete Lagrangian formulation quantifying the irreversible cost of world construction. This section develops the mechanical framework and connects it to the RSVP field theory.

8.1 Optionality and the Discrete Lagrangian

Let $\Omega(X) \in \mathbb{R}_{\geq 0}$ denote the *optionality* of an option-space X: a monotone measure satisfying $X' \subseteq X \Rightarrow \Omega(X') \leq \Omega(X)$. Each event $e_t : X_{t-1} \to X_t$ induces a local cost

$$L_t = \Delta\Omega_t + \lambda\, \Delta C_t, \tag{10}$$

where $\Delta\Omega_t = \Omega(X_{t-1}) - \Omega(X_t)$ is the reduction in optionality and ΔC_t records auxiliary accounting costs (ethical, normative, or thermodynamic), with λ the relative weighting constant.

The *action* of a history $\gamma = (e_1, \ldots, e_n)$ is

$$S[\gamma] = \sum_{t=1}^{n} L_t. \tag{11}$$

Action is monotone increasing along histories for Pop, Ref, and Bind, encoding irreversibility mechanically.

8.2 Commitment and the Hamiltonian

Define the conjugate of optionality as

$$\pi_t = -\frac{\partial L_t}{\partial \Omega}, \tag{12}$$

interpreted as *commitment*: the marginal cost of further optionality loss. The discrete Hamiltonian is

$$H_t = \pi_t \, \Omega(X_t) - L_t, \tag{13}$$

measuring *remaining freedom*. H_t decays as commitment accumulates; a system with $H_t \approx 0$ is fully world-bound.

This formulation inverts optimization: the goal is not to minimize action but to *account for* it. Meaningful histories have non-trivial action; systems that avoid commitment remain vacuous.

8.3 Collapse and Action Reduction

Collapse is the only operator that may reduce accumulated action. For a policy q inducing $\sim_q$ with kernel $\ker q$:

$$S \longmapsto S - \log |\ker q|. \tag{14}$$

This corresponds, under F, to the suppression of high-frequency modes in the RG flow image $F(\mathrm{Col})$: fine-structure entropy is integrated out, reducing the action by exactly the log of the number of identified degrees of freedom.

8.4 Connection to RSVP Dynamics

The discrete equations of motion

$$X_{t+1} = e_{t+1}(X_t), \qquad \pi_{t+1} - \pi_t = -\Delta\Omega_{t+1}$$

correspond, under F, to the integral form of (5): the entropy transport equation is the continuous limit of the discrete commitment-optionality duality. The diffusion coefficient κ in RSVP corresponds to the rate at which boundary-sharpening from $F(\mathrm{Pop})$ propagates across M.

9 Sheaf-Theoretic Interpretation of Worldhood and Meld

9.1 Local Scopes and the Presheaf of Proposals

An option-space X admits a cover $\mathcal{U} = \{U_i\}$ of local scopes (semantic regions, rhetorical roles, subproblems). Define a presheaf $\mathcal{T}$ assigning to each open $U \subseteq X$ the set of admissible local update proposals:

$$\mathcal{T}(U) = \{\text{local tangent proposals valid on } U\},$$

with restriction maps $\rho_{UV} : \mathcal{T}(U) \to \mathcal{T}(V)$ for $V \subseteq U$. A family $\{\delta_i \in \mathcal{T}(U_i)\}$ is *compatible* if on each overlap:

$$\rho_{U_i,U_i\cap U_j}(\delta_i) = \rho_{U_j,U_i\cap U_j}(\delta_j).$$

9.2 Commitment as Gluing

Commitment resolves local incompatibilities. When an agent enforces compatibility through Pop, Bind, or Ref, local sections are constrained so that they may glue. The sheaf condition—compatible local sections determine a unique global section—corresponds precisely to the achievement of *worldhood*: local possibilities resolved into a coherent whole.

Collapse admits a direct sheaf-theoretic interpretation as *sheafification*: the minimal modification of a presheaf enforcing the sheaf condition by quotienting incompatible distinctions. Sheafification = collapse = abstraction.

9.3 Meld as Policy-Induced Sheafification

The following theorem formalizes the main result of the geometric treatment:

Theorem 9.1 (Meld as sheafification)**.** *Let $\mathcal{T}$ be a presheaf of local proposals over cover $\mathcal{U}$, and let $\{\delta_i \in \mathcal{T}(U_i)\}$ fail to glue due to overlap obstructions. Fix a merge policy π specifying an identification relation $\sim_\pi$ on the obstruction data. Then there exists a canonical sheafification $\mathcal{T} \Rightarrow \mathcal{T}^\pi$ in which the δ_i become compatible and glue uniquely to a global section $\Delta^\pi \in \mathcal{T}^\pi(X)$. The Spherepop operation* Meld_π *is precisely the event that commits to this identification and takes Δ^π as authoritative continuation.*

Moreover, the decomposition $\mathrm{Meld}_\pi \equiv \mathrm{glue} \circ \mathrm{Col}_\pi$ (Proposition 3.4) is exactly the sheafification decomposition: collapse removes the obstructions; gluing produces the unique global section.

9.4 Minimal Commitment and Maximal Safe Gluing

Minimal commitment paths are characterized by the principle: *glue as much as possible; collapse only as necessary; defer the rest.* Formally, among all policies π and subcovers $\mathcal{U}' \subseteq \mathcal{U}$ that reach a target semantic class $[D]$, those that maximize the glued domain $\bigcup \mathcal{U}'$ while minimizing policy-induced identification $|\ker \pi|$ yield histories with minimal irreversible action. This follows because gluing increases coherence without increasing irreversible identification, while collapse directly contributes to action cost via (14).

9.5 Document Construction and Semantic Manifolds

In the document construction setting, local scopes U_i correspond to rhetorical regions (definitions, pivots, proofs); overlaps are shared commitments (terminology, notation, narrative voice). The chart-invariant tangent flow result can be summarized: a generative program that selects weighted local proposals δ_i with weights $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$, enforcing overlap compatibility, realizes a minimal commitment step toward any reachable target semantic class. Concretely:

$$\Delta = \mathrm{glue}\left(\sum_{i \in A} \alpha_i \, \delta_i\right) \in \mathcal{T}^\pi(X),$$

and this weighted local-gluing representation is equivalent, under F, to a tangent-constrained semantic flow on $M = \Delta(\Omega)$.

10 Mathematical Status and Open Problems

10.1 Fully Formalized

The following components are rigorous and reducible to standard results:

1. The category **SP**: reduces to rewriting systems and quotient categories over posets with entropy measures.

2. The monoidal structure (Proposition 3.2): standard once generators and composition law are fixed.

3. The universal property of Col (Proposition 3.3): standard coequalizer argument.

4. Meld as pushout (Proposition 3.4): standard pushout construction in **SP**.

5. Functoriality of F (Theorem 6.3): object and morphism maps are given explicitly; functor laws verified from locality of perturbations.

6. SPC type safety (Preservation + Progress) by induction on typing derivations.

7. Adequacy of probabilistic semantics: printed via the Giry monad laws.

8. Meld as sheafification (Theorem 9.1): follows from the sheafification universal property [Mac Lane & Moerdijk, 1992].

10.2 Modeling-Dependent

1. The choice of $\Delta(\Omega)$ as the smooth realization of option-spaces. Alternative realizations (Wasserstein spaces, configuration manifolds) are possible.

2. The specific form of $\delta\Phi_\partial$ in $F(\mathrm{Pop})$. Existence is guaranteed; the precise profile depends on the metric on M.

3. The entropy-transport equation (5). Alternatives (Fokker–Planck, heat equation) produce different RG flows under $F(\mathrm{Col})$.

10.3 Open Problems

1. **Faithfulness of F.** Is F faithful—do distinct morphisms in **SP** always produce distinct smooth maps in **RSVP**? Likely requires additional structure on $\Delta(\Omega)$ (e.g., a non-degenerate Riemannian metric).

2. **Non-trivial entropy for Bind.** Constructing an entropy functional on **SP** that makes Axiom 2 strictly non-trivial for Bind alone (which preserves Shannon entropy globally while reducing symmetry) requires a finer invariant—possibly topological entropy on the admissibility lattice $\mathcal{A}$, or a second-order entropy measuring the complexity of the dependency graph.

3. **Compact closed completion.** Does **SP** admit a completion to a compact closed or †-compact category? This would enable a graphical calculus for Spherepop diagrams and connect the framework to string-diagram methods in quantum foundations.

4. **Strong normalization.** SPC enjoys strong normalization for the pure deterministic fragment (inherited from CoC). The conjecture that SPC has probabilistic normalization almost surely—as in Kozen-style probabilistic λ-calculi—remains open.

5. **π-calculus fragment.** The translation T_π into SPC handles communication-free parallelism but not full name-passing mobility. Whether a name-passing extension of **SP** exists that remains entropy-decreasing is open.

11 Conclusion

Compression. The content of this paper compresses to one transportable sentence:

> *Spherepop is a finitely generated entropy-decreasing rewriting category whose morphisms—option-elimination, dependency binding, and quotient collapse—are realized geometrically by the RSVP functor as boundary sharpening, directed vector-field coupling, and renormalization, with the deepest structural asymmetry being that Spherepop accumulates constraint while RSVP redistributes coherence.*

The four-way unification. The paper demonstrates that what appeared as four independent trajectories—event-history calculus, typed functional language, OS kernel semantics, and geometric field theory—are in fact four presentations of a single mathematical object. The event-history calculus provides the axiomatic core. SPC provides the type-theoretic instantiation in a presheaf topos. The OS kernel provides the operational/systems instantiation. RSVP provides the analytic completion via the functor F.

Deepest asymmetry. Spherepop *accumulates*: each morphism adds an irreversible record to the log, and the state is always a residue of that accumulation. RSVP *redistributes*: the PDE (5) diffuses local entropy gradients across M, and global coherence is a time-averaged field property. One is combinatorial and log-bound; the other is differential and field-bound. The functor F mediates between them without collapsing this difference—and it is precisely this preserved asymmetry that makes the functor mathematically non-trivial.

Joy. In a universe of infinite options, nothing matters. It is only by popping futures, refusing paths, binding promises, and collapsing detail that a world becomes livable. The joy of Spherepop lies here: not in maximal openness, but in the relief of resolution—the moment when possibility condenses into reality, and a finite path forward appears.

A Notation Summary

Symbol	Meaning
SP	Category of Spherepop histories
RSVP	Category of RSVP states and entropy-flux maps
Ω	Option-space (object of **SP**)
$\mathcal{A}$	Admissibility family on Ω
H	Entropy functional $\mathrm{Ob}(\mathbf{SP}) \to \mathbb{R}_{\geq 0}$
$\mathrm{Pop}, \mathrm{Ref}, \mathrm{Bind}, \mathrm{Col}$	Four primitive generating morphisms
Meld_π	Pushout of histories under policy π
F	Geometric realization functor $\mathbf{SP} \to \mathbf{RSVP}$
Φ	Scalar coherence potential on M
v	Velocity vector field on M
S	Entropy density on M
κ	Diffusion coefficient in (5)
$\partial_P \Phi$	Pop derivative (6)
μ°	Merge product (coherence-preserving convolution)
K	Kolmogorov complexity
$\sim_q$	Collapse-policy equivalence on histories
$\Delta(\Omega)$	Probability simplex over Ω
$\Omega(X)$	Optionality measure on option-space X
$S[\gamma]$	Action of history γ
π_t	Commitment (conjugate to optionality)
H_t	Hamiltonian (remaining freedom)
$\mathcal{T}$	Presheaf of local semantic proposals
Dist	Giry distribution monad

B SPC Grammar and Typing Rules (Summary)

Grammar.

$$t, u \ ::= \ x \ \mid \ a \ \mid \ \mathrm{Sphere}(x\!:\!A.\,t) \ \mid \ \mathrm{Pop}(t,u) \ \mid \ \mathrm{Merge}(t,u) \ \mid \ \mathrm{Choice}(p,t,u)$$

Selected Typing Rules.

$$\frac{\Gamma, x\!:\!A \vdash t : B}{\Gamma \vdash \mathrm{Sphere}(x\!:\!A.\,t) : \Pi x\!:\!A.\,B} \qquad \frac{\Gamma \vdash f : \Pi x\!:\!A.\,B \quad \Gamma \vdash u : A}{\Gamma \vdash \mathrm{Pop}(f,u) : B[u/x]}$$

$$\frac{\Gamma \vdash t : A \quad \Gamma \vdash u : A}{\Gamma \vdash \mathrm{Merge}(t,u) : A} \qquad \frac{\Gamma \vdash p : \mathrm{Prob} \quad \Gamma \vdash t : A \quad \Gamma \vdash u : A}{\Gamma \vdash \mathrm{Choice}(p,t,u) : A}$$

Reduction Rules.

$$\mathrm{Pop}(\mathrm{Sphere}(x\!:\!A.\,t), u) \ \to \ t[u/x] \qquad \mathrm{Merge}(t,u) \ \equiv \ \mathrm{Merge}(u,t)$$

$$\mathrm{Choice}(p,t,u) \ \to \ \begin{cases} t & \text{w.p. } p \\ u & \text{w.p. } 1-p \end{cases}$$

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